Mathematics & Advance Statistics

PR-1 : Expectation Decider

* Understanding the Basics:

1. What is Probability?
- Probability measures the chance of an event happening
- It is between 0 and 1.
- Probability = 0 ⇒ event is impossible
- Probability = 1 ⇒ event is certain.

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total Numbers of outcomes}}$$

2. Key probability Terminology:

o Experiment:
- An experiment is an action that produces Outcomes.

o Outcome:
- A single possible result of an experiment.

o Sample Space (S):
- The set of all possible outcomes.

o Event:
- An event is one or more outcomes from the sample space.

o Favourable Outcomes:
- Outcomes that match the desired event.

Example:

Experiment: Selecting a student from the dataset.

Outcome : The selected student passes the exam.

Sample Space : {Pass, fail}

Event : Student attended more than 80%

3. Three probability Event examples from the dataset.

- Event 1:

  A student studies more than 10 hrs / week.

- Event 2:

  A student has attendence more than 80%

- Event 3:

  A student participates in group discussion and passes the final exam.

* Types of Probability:

1. Theoretical Probability:

- Calculated using methematical reasoning & formulas.

$$P(E) = \frac{\text{Favourable Outcomes}}{\text{Total Possible Outcomes}}$$

Example:

- Suppose student is equally likely to pass or fail.

$$P(Pass) = \frac{1}{2} = 0.5$$

2. Empirical Probability:

- Calculated using actual observed data from experiment

$$P(E) = \frac{\text{Number of time event Occurs}}{\text{Total Observations}}$$

Example:

Suppose 150 out of 200 students passed.

$$P(Pass) = \frac{150}{200} = 0.75$$

* Random Variable & Probability Distribution

1. Define a random variable for the event "Number of students passing the final exam" out of 3 randomly selected students.

→ Let X = Number of students passing among 3 randomly Selected students.

$X = \{0, 1, 2, 3\}$

- In this case X is random variable which result is vary for different experiments.

2. Construct it's probability distribution table.

Assume P(Pass) = 0.75

$$P(X) = \binom{n}{x} p^x (1-p)^{n-x}.$$

$n = 3$, $P = 0.75$

$\Rightarrow P(X=0) = \binom{3}{0} (0.75)^0 \cdot (0.25)^3$

$= 1 \times 1 \times 0.01562$

$= \underline{0.016}$

$\Rightarrow P(X=1) = \binom{3}{1} (0.75)^1 \cdot (0.25)^2$

$= 3 \times 0.75 \times 0.0625$

$= \underline{0.1406}$

$\Rightarrow P(X=2) = \binom{3}{2} (0.75)^2 \cdot (0.25)^1$

$= 3 \times 0.5625 \times 0.25$

$= \underline{0.4218}$

$\Rightarrow P(X=3) = \binom{3}{3} \times (0.75)^3 \cdot (0.25)^0$

$= 1 \times 0.4218$

$= \underline{0.4218}$

X	0	1	2	3
P(X)	0.016	0.141	0.422	0.422

3. find mean & variance:

$\Rightarrow$ Expected Mean:

$E(X) = \sum x.P(x)$

$= 0 \times 0.016 + 1 \times 0.1406 + 2 \times 0.428 + 3 \times 0.4218$

$= \underline{2.25}$

$$Var(X) = \sum (x_i - \mu)^2 \cdot P(x_i)$$
$$= [(0-2.25)^2 \times 0.016] + [(1-2.25)^2 \times 0.1406]$$
$$+ [(2-2.25)^2 \times 0.4218] + [(3-2.25)^2 \times 0.4218]$$
$$= 0.081 + 0.2196 + 0.02636 + \cancel{0.659}\ 0.2373$$
$$= \cancel{1.29}\ \underline{0.56}$$

OR

Use can use simple formulas like

$E(X) = np = 3 \times 0.75 = \underline{\underline{2.25}}$

$Var(X) = np(1-p) = 3 \times 0.75 \times 0.25 = \underline{\underline{0.56}}$

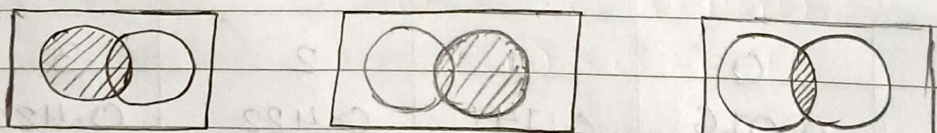
* Venn Diagram in Probability:

~~A=~~ Draw Venn Diagram of:

Suppose,

A = Students study more than 10 hrs / week

B = students attend more than 80% of classes.



A B $A \cap B$

* Contigency Table & Probability Calculations.

1. Create a Contigency table for group-discussion (Yes/No) vs. final-exam-pass (Pass/fail)

	Pass	fail	Total
Yes	90	20	110
NO	60	30	90
Total	150	50	200

2. Find Joint probability of "Participates in group discussion AND Passes exam."

$P(\text{Pass} \cap \text{Yes}) = \frac{90}{200} = 0.45$

3. Marginal Probability of Passing Exam.

$P(\text{Pass}) = \frac{150}{200} = 0.75$

4. Conditional Probability of "Passing exam given participation in group discussion".

$$P(\text{Pass} \mid \text{Yes}) = \frac{P(\text{Pass} \cap \text{Yes})}{P(\text{Yes})}$$

$$= \frac{90/200}{110/200}$$

$$= 0.8182$$

* Understanding Relationships

1. Interpret Conditional Probability result in plain language
- In our dataset P(Pass | Yes) is high. It mean students who participate in group discussions have a greater chance of passing the exam

2. Identify if "Participating in group discussions" and "Passing exam" are independent, dependent or mutually exclusive events. Justify

→ Group discussion and passing the exam are dependent events because participating in discussion can improve understanding and increase the probability of passing.

* Bayes Theorem Application:

~~P(Pass | high attendance) = 0.7~~

P(High attendance | Pass) = 0.7
P(High attendance | Fail) = 0.4
P(High attendance) = 0.6

~~P(Pass) =~~
P(High attendance) = P(H|P) × P(P) + P(H|F) × P(F)
0.6 = 0.7 × P(P) + 0.4 (1 − P(P))
(∵ P(P) + P(F) = 1)
∴ 0.6 − 0.4 = 0.7 × P(P) − 0.4 P(P)
∴ 0.2 = 0.3 × P(P)
∴ P(P) = 0.6667

$$\therefore P(\text{Pass} \mid \text{High attendance}) = \frac{P(H \mid P) \cdot P(P)}{P(H)}$$
$$= \frac{0.7 \times 0.67}{0.6}$$
$$= 0.7778$$

- This indicates that students with high attendance have higher chance of passing the final exam.